KIAN ENTEZARI

Computer Vision | Embedded Systems | Robotics | Biosensors

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SUMMARY

Electrical and Computer Engineering student at UC Berkeley with a 4.00 GPA and hands-on experience in computer vision, embedded systems, and biosensor development. Built a stereo vision pipeline for a commercial robotic arm at Mantis Robotics and conducting independent research on a low-cost electrochemical potentiostat for non-invasive sweat analysis.

EDUCATION

University of California, Berkeley *Expected Spring 2028*
B.S., Electrical and Computer Engineering | GPA: 4.00

TECHNICAL SKILLS

Languages: Python, C, C++, C#
Perception: DIAVision, OpenCV, YOLOv8, LiDAR
Electronics: KiCad, LTspice, PCB Design
Machine Learning: PyTorch, ONNX Runtime
Tools: AutoCAD, Git, ROS 2
Protocols: Modbus TCP, Google API, Meta API, OpenAI API
Spoken Languages: Spanish (conversational)
Embedded Platforms: ESP32, Arduino

EXPERIENCE

Computer Vision Engineer Intern | Mantis Robotics *Nov 2025 – Aug 2026*

- Developed a stereo vision pipeline for the MR-1 robotic arm achieving multi-object 3D localization with <1 mm accuracy, while cutting hardware costs by over 90% by replacing industrial cameras with two low-cost webcams.
- Integrated YOLOv8 object detection (OpenCV, ONNX Runtime) for real-time object detection and class identification.
- Implemented camera-to-robot coordinate transformations for robotic object localization.
- Built a LiDAR-camera fusion pipeline using YOLO detections and projected point clouds for 3D object localization.
- Created a Modbus TCP server to transmit robot coordinates from the vision pipeline to the robot controller.
- Integrated and tested digital I/O for robotic hardware communication and control.

Software Engineer Intern | Inner Health & Wellness *Oct 2025 – Jan 2026*

- Built and deployed an AI chatbot into the company's app using the OpenAI API, giving clients an in-app resource for health and wellness guidance.
- Developed a backend server and data ingestion workflow using the Google API to centralize company content and deliver it in-app.
- Integrated the Meta API for in-app messaging and deployed all services on Vercel.

Tutor | Mathematics Tutoring *Jan 2026 – Present*

- Tutor Calculus through Differential Equations, adapting to individual learning styles and pacing.

PROJECTS

Independent Research | Multi-Analyte Electrochemical Biosensor *May 2026 – Present*

- Designing and building a low-cost TLV272-based potentiostat circuit capable of measuring blood sugar, lactate, and other analytes (cortisol, blood alcohol, vitamin C) from sweat via electrochemical detection.
- Developing a reverse iontophoresis system for non-invasive interstitial fluid extraction, enabling continuous monitoring without skin puncture.